

A Practitioner's Guide to Inference with Spatial Dependence

Stefano DellaVigna
UC Berkeley
sdellavi@econ.berkeley.edu

Guido Imbens
Graduate School of Business
Stanford University
imbens@stanford.edu

Woojin Kim
Stanford University
wjnkim@stanford.edu

David M. Ritzwoller
Stanford University
ritzvoll@stanford.edu

Damian Clarke
Department of Economics
University of Chile
dclarke@fen.uchile.cl

Nicolás París
Department of Economics
University of Chile
nparis@fen.uchile.cl

Abstract. In many empirical applications in the social sciences, individuals-level observations interact or are subject to shared shocks across space. In order to conduct valid statistical inference, such spatial dependence must be accounted for. While a number of strategies to account for spatial dependence exist, these typically are based on strict assumptions about delimitation of dependence within certain geographic areas (*i.e.* clustering), or modelling dependence solely in terms of distance. In this paper, we discuss these common forms of inference with spatial dependence and their drawbacks, and introduce an alternative form of inference, laid out in [DellaVigna et al. \(2025\)](#), known as Thresholding Multiple Outcomes (TMO). This procedure for inference models dependence by using information from a range of variables to infer the dependence between units across space, and constructs standard errors and confidence intervals based on such dependence. We introduce the command `tmo`, which implements these procedures in Stata for a range of linear models including OLS and 2SLS.

Keywords: spatial dependence, inference, multiple outcomes, standard errors.

1 Introduction

Spatial dependence is a common feature of data in the social sciences. Interactions between individuals in their neighbourhoods, cities, states or other social networks typically imply that individuals cannot be viewed as independent observations in space. And even absent direct interactions between individuals, shared underlying shocks typically occur in areas which are linked across potentially quite broad areas of space, and potentially in quite a complex fashion. Productivity shocks in one region propagate to others through trade and labour-market linkages, policies adopted in one jurisdiction shape both policy adoption and outcomes elsewhere, and common shocks such as weather events, commodity price movements and health crises strike collections of locations that need not be contiguous. Importantly, the locations which co-move need not be near neighbours: two distant, but economically similar, agricultural counties may respond to a commodity price shock far more similarly than two adjacent counties, one urban and one rural.

Such spatial dependence must be accounted for if one wishes to conduct inference on parameters estimated in econometric models. Variance-covariance estimators in commonly estimated econometric models such as ordinary least squares (OLS) and two stage least squares (2SLS) typically start from the assumption of independence, but must be relaxed to account for more realistic interactions among observations across space. If such adjustments are not made, standard errors will be mis-estimated, generally resulting in standard errors which are too small, and confidence intervals which are too narrow.

In this paper we lay out considerations for conducting inference in models where spatial dependence among observations is suspected. We discuss a number of existing common procedures and their limitations, and lay out a new procedure for inference with spatial correlation variance which leverages multiple outcome measures. This procedure, referred to as Thresholding Multiple Outcomes (TMO) is described in [DellaVigna et al. \(2025\)](#) (and in this paper), and is accompanied by a Stata routine `tmo` designed for a range of commonly encountered estimation methods in Stata which we document here.

Tools to relax the variance-covariance matrix *in certain ways* are default in many computational implementations in Stata. For example, essentially all Stata estimators incorporate heteroscedasticity-robust variance estimators. Similarly, cluster-robust variance estimation allows for *limited* spatial dependence: observations may correlate arbitrarily within a designated group, a state for instance, but are assumed independent across groups. The consequences of ignoring such intra-group correlation have long been understood ([Moulton 1986, 1990](#)), and clustering is now routine empirical practice.

However, clustering assumes a very specific form of dependence, in which correlation stops abruptly at the boundary of the cluster. A second family of procedures instead models dependence explicitly as a function of distance, either geographic or ‘economic’. [Conley \(1999\)](#) proposes a kernel-based estimator in which the covariance between two locations is down-weighted as the distance between them grows, reaching zero at a chosen cutoff. More recently, [Müller and Watson \(2022b,a\)](#) propose Spatial Correlation

Principal Components (SCPC), which conducts inference using a small number of low-frequency principal components derived from a worst-case spatial correlation model, offering improved size control when observations agglomerate in space.

Both families, however, impose *a priori* that dependence is organised by geography: either delimited by borders, or declining in distance. It is reasonable to ask what dependence looks like in practice. DellaVigna et al. (2025) propose to answer this question empirically: when a researcher observes many outcomes for the same units, as is very often the case, those auxiliary outcomes can be used to *estimate* which pairs of locations are correlated, rather than assuming where correlation lies. Their procedure, Thresholding Multiple Outcomes (TMO), then conducts inference on the parameter of interest allowing for dependence among precisely those pairs.

To see how these approaches differ, consider a representative county: Tulsa, Oklahoma. Figure 1 displays, for each of the four procedures discussed above, which other counties enter the variance calculation for a regression using county-level data, and with what weight. Clustering by state (panel a) admits arbitrary correlation between Tulsa and the rest of Oklahoma, and nothing else. The Conley approach (panel b) weights counties by a declining function of their distance to Tulsa, reaching zero at 150 miles: dependence is assumed strongest nearby, and absent far away. SCPC (panel c) is displayed through its implicit pair-level kernel: because inference is based on a handful of smooth, low-frequency spatial components, *every* county in the sample receives some weight, largest in a broad region around Tulsa and varying smoothly over the whole map, with no threshold and no cutoff. TMO (panel d) is the mirror image: using 90 auxiliary county-level outcomes, it identifies the small set of counties whose outcomes systematically co-move with Tulsa's (about 0.5% of all pairs sample-wide), and allows arbitrary correlation with those counties only, wherever they happen to be on the map, including far beyond any distance cutoff.

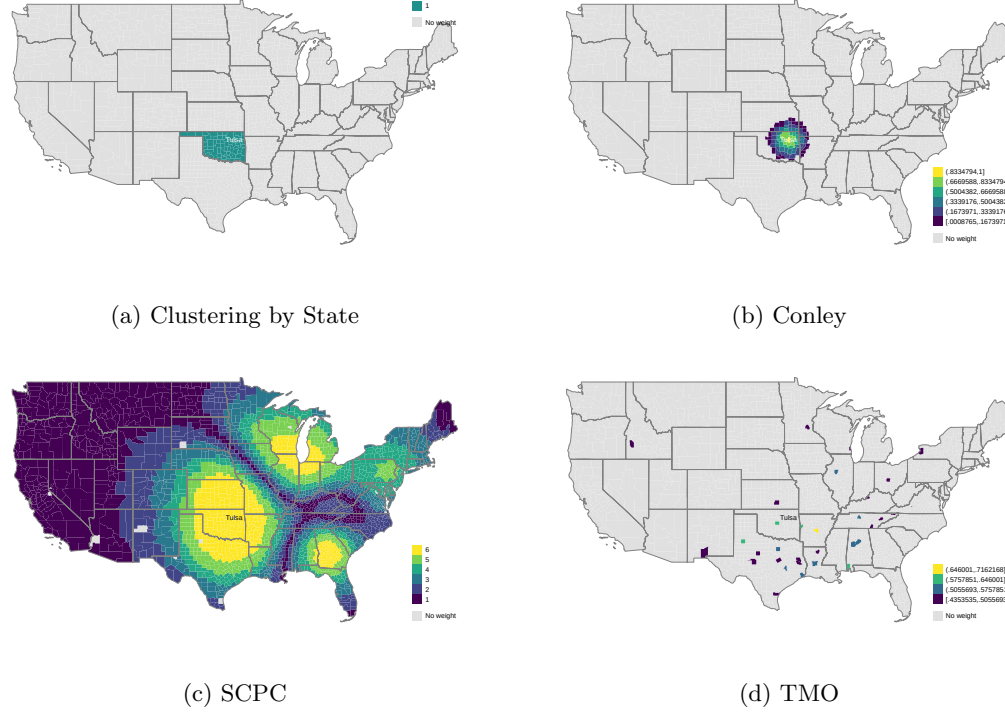


Figure 1: Spatial Standard Error Procedures

In this paper we provide a primer on estimating standard errors under spatial dependence in Stata, introduce the TMO procedure of DellaVigna et al. (2025) and its implementation in the `tmo` command, and work through a series of examples based on a rich collection of U.S. county-level outcomes. A distinguishing feature of the command is that TMO need not displace existing practice: it can *augment* clustered, distance-based, or SCPC inference, adding to the variance calculation those pairs of locations whose auxiliary outcomes reveal dependence that the baseline procedure would miss. We document these combinations, alongside cross-sectional, instrumental variables, and panel implementations.

A number of related tools exist for spatial inference in Stata, including implementations of Conley standard errors, the `scpc` command of Müller and Watson, and the `spur` suite of Becker et al. (2026) for diagnosing spatial unit roots. The `tmo` command complements these by providing inference based on estimated, rather than assumed, patterns of cross-location dependence.

In what remains of this paper, we begin by describing methods in Section 2, first discussing proposals to deal with spatial dependence, before focusing on TMO more generally. We discuss a Stata command to implement TMO in Section 3, before docu-

menting a number of examples of the use of this command, `tmo`, in Section 4. Section 5 lays out key points of relevance for practitioners and concludes.

2 Methods

2.1 A Simple Bivariate Regression Model

Consider estimating a simple bivariate regression model of the type:

$$Y_i = \alpha + \tau W_i + \varepsilon_i, \quad (1)$$

based on a sample of data $i = 1, \dots, N$, where Y_i is some outcome of interest, W_i some treatment or independent variable of interest and ε_i an unobserved term. Assume now that $\mathbb{E}[\varepsilon_i | W_i] = 0$. In this setting, the parameters α and τ can be estimated consistently by ordinary least squares, and denoted $\hat{\alpha}$ and $\hat{\tau}$ respectively. Here, as in most empirical settings, we are most interested in the quantity $\hat{\tau}$.

While the estimation of $\hat{\tau}$ is straightforward, the estimation of the standard error for this estimate hinges upon assumptions about the correlation among residuals ε_i within the sample. To see this, note that the standard error is the square root of the following variance term:

$$\mathbb{V}(\hat{\tau} | W) = (W'W)^{-1} W' \Omega W (W'W)^{-1}. \quad (2)$$

This depends, crucially, on the matrix Ω which is an $N \times N$ matrix describing the covariance between each ε_i in the sample. The most restrictive assumption about this matrix is that all observations share a common variance, and all observations are independent, or in other words, observations are independent and identically distributed. In this case, Ω simplifies to $\sigma^2 I$, with σ^2 referring to the variance of Y_i , and I and $N \times N$ identity matrix. This leads to the simple variance expression:

$$\mathbb{V}(\hat{\tau} | W) = \sigma^2 (W'W)^{-1}, \quad (3)$$

which is returned as default in computational estimators of (1).

The assumption of a common variance can be loosened quite simply to reflect heteroscedasticity, as laid out in seminal work of [Huber \(1967\)](#); [Eicker \(1967\)](#); [White \(1980\)](#). In this case, rather than (3), the heteroscedasticity-robust analogue:

$$\hat{\mathbb{V}}(\hat{\tau} | W)_H = (W'W)^{-1} \left(\sum_{i=1}^N \hat{\varepsilon}_i^2 W_i W_i' \right) (W'W)^{-1} \quad (4)$$

can be estimated, where $\hat{\varepsilon}_i = Y_i - \hat{\alpha} - \hat{\tau} W_i$. Again, such a variance estimator is standard in most computational implementations of regression estimators, such as Stata's `robust` option.¹ However, both (3) and (4) are predicated on the assumption that observations are independent, and as such ε_i and ε_j are not correlated for all observation pairs i and

¹In practice, because of finite sample biases in this estimator, most software packages report finite-sample corrected versions of the heteroscedasticity-robust variance estimator. Stata's `robust` option

j where $i \neq j$. In many cases, this is likely viewed as an unrealistic assumption, and instead some sort of spatial (or other) correlation is presumed to exist between terms ε_i and ε_j . In this case, a number of procedures exist to seek to conduct appropriate inference despite such dependence.

Restricting Elements of Ω

A first way to proceed is to note that we can permit limited sorts of spatial dependence by clustering at some aggregate geographic level. This recognises the importance of intra-group clustering (Moulton 1986), by allowing certain dependence across space within Ω while restricting elements of Ω outside of any specific cluster to be 0. In essence, this allows arbitrary correlations within some specific geographical or other boundary, while restricting any dependence across clusters. Such a case can be viewed schematically in Ω_{CR} below, for a cluster-robust case assuming a number of blocks, compared to the more restrictive heteroscedasticity-robust option Ω_H .

$$\Omega_{CR} = \begin{bmatrix} \varepsilon_1^2 & \varepsilon_1\varepsilon_2 & 0 & 0 & \cdots & 0 & 0 \\ \varepsilon_2\varepsilon_1 & \varepsilon_2^2 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \varepsilon_3^2 & \varepsilon_3\varepsilon_4 & \cdots & 0 & 0 \\ 0 & 0 & \varepsilon_4\varepsilon_3 & \varepsilon_4^2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \varepsilon_{N-1}^2 & \varepsilon_{N-1}\varepsilon_N \\ 0 & 0 & 0 & 0 & \cdots & \varepsilon_N\varepsilon_{N-1} & \varepsilon_N^2 \end{bmatrix} \quad \Omega_H = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}$$

Inference based on clustering is standard, following Liang and Zeger (1986); MacKinnon and White (1985),

$$\widehat{V}(\widehat{\tau} | W)_{CR} = (W'W)^{-1} \left(\sum_{g=1}^G W_g' \widehat{\varepsilon}_g \widehat{\varepsilon}_g' W_g \right) (W'W)^{-1} \quad (5)$$

where the g refers to each of the G clusters, $\widehat{\varepsilon}_g$ a vector of residuals for each cluster. In practice, Stata's implementation of such cluster-robust variance estimates incorporates a finite sample correction factor, multiplying (5) by $\frac{G}{G-1} \frac{N-1}{N-K}$ to correct for the downward bias in standard errors when the number of clusters is small (Cameron and Miller 2015; Rogers 1993).

Distance-based limiting

A drawback of such an approach is that spatial dependence is viewed as being sharply demarcated by the borders or limits of clusters. Another common alternative which

reports the HC1 estimator by default, which rescales the HC0 matrix in (4) by $\frac{N}{N-k}$, where k is the number of estimated parameters (2 in this case). Other common variants include HC2 and HC3, which apply leverage-based corrections to adjust for influential observations (see e.g. MacKinnon and White 1985; Long and Ervin 2000), also available via Stata's `vce()` option.

seeks to avoid such sharp demarcation at geographic borders instead views spatial dependence in terms of distance. This approach, proposed by [Conley \(1999\)](#), allows for residual dependence to decay continuously over space. Rather than imposing a block-diagonal structure in Ω , Conley’s estimator permits non-zero covariances between all observations but downweights them by a kernel function of geographic distance, up to some maximum point at which case zero covariance is assumed.

Formally, the Conley variance estimator takes the form:

$$\widehat{\mathbb{V}}(\hat{\tau} | W)_{\text{Conley}} = (W'W)^{-1} \left(\sum_{i=1}^N \sum_{j=1}^N K_{ij} \hat{\varepsilon}_i \hat{\varepsilon}_j W_i W_j' \right) (W'W)^{-1}, \quad (6)$$

where $K_{ij} = K(d_{ij}/b)$ is a kernel function of the distance d_{ij} between units i and j , and b is a bandwidth parameter. Typical choices for $K(\cdot)$ include the Bartlett or uniform kernel.² Intuitively, this estimator treats spatial correlation as strongest between nearby units, either uniformly, or declining smoothly with distance, and ultimately reaching zero beyond a certain cutoff. We can see intuitively that as K_{ij} decays to zero, the correlation between residuals in (6) becomes irrelevant in the variance estimate, as the summation term resolves to 0 when $K_{ij} = 0$.

There are also alternative approaches which seek to take such-distance based approaches. [Müller and Watson \(2022b\)](#) point out that if observations are not uniform in space, then methods such as (6) will not be asymptotically valid, and potentially suffer from size distortions in finite samples which lead to over-rejection of null hypotheses. Thus, this leads to suggestions as laid out in [Müller and Watson \(2022b,a\)](#) in which rather than using kernel functions to model the location of units across space, the location of units is modelled based on principal components which capture spatial agglomeration. They show that this approach offers improved performance in cases where spatial agglomeration of observations occurs, and, when confidence intervals are constructed based on critical values which consider such spatial dependence, can result in large improvements in error rates ([Müller and Watson 2022a](#)).

Thresholding Multiple Outcomes

Using Multiple Outcomes to Infer Cross-Space Correlations Ultimately, however, both cluster-robust and distance-based procedures are limiting, either by assuming a strict off-diagonal zero correlation, or by assuming a zero correlation among units beyond some specified cut-off point. This limitation occurs given that these approaches are based entirely upon modelling spatial correlations by considering how *units* relate across space, rather than considering how outcomes Y_i themselves relate across space. Thus, a more realistic approach may be to seek to infer geographic dependence by examining the correlation of a range of actual measures over space.

²These are defined respectively as $K(d_{ij}/b) = 1 - d_{ij}/b$ for $d_{ij} \leq b$ and 0 otherwise, which applies linearly decreasing weights as distance increases; and $K(d_{ij}/b) = \mathbb{1}\{d_{ij} \leq b\}$, which applies equal weight to all pairs within a cutoff distance b .

To this end, DellaVigna et al. (2025) introduce Thresholding Multiple Outcomes, or TMO, in which auxiliary measures are used to determine whether correlations exist between particular locations in data. This requires information on a number d of auxiliary outcomes, which we will denote $Y_i^{(1)}, \dots, Y_i^{(d)}$. If such multiple alternative outcomes exist, we can use these measures to assess whether systematic correlation exists between residuals over space.

Namely, each of these outcomes which we will generically refer to as $Y_i^{(j)}$ can themselves be regressed on our outcome of interest in (1). The residual term from this regression is denoted $\hat{\varepsilon}_i^{(j)}$, and this can be normalised to have a variance of 1 as $\bar{\varepsilon}_i^{(j)} = \hat{\varepsilon}_i^{(j)} / \sqrt{\frac{1}{N}(\hat{\varepsilon}_i^{(j)})^2}$. We will seek to leverage these residuals to determine the correlation of outcomes between locations across space. To do so, the covariance and the correlation between residuals for some locations i and i' are denoted:

$$\hat{\lambda}_{i,i'} = \frac{1}{d} \sum_{j=1}^d (\bar{\varepsilon}_i^{(j)} - \bar{\varepsilon}_i)(\bar{\varepsilon}_{i'}^{(j)} - \bar{\varepsilon}_{i'}) \quad \text{and} \quad \hat{\rho}_{i,i'} = \frac{\hat{\lambda}_{i,i'}}{\sqrt{\hat{\lambda}_{i,i} \hat{\lambda}_{i',i'}}}, \quad (7)$$

where $\bar{\varepsilon}_i = \frac{1}{d} \sum_{j=1}^d \bar{\varepsilon}_i^{(j)}$. This quantity $\hat{\rho}_{i,i'}$ is of key importance, as it summaries, across all outcomes d , the tendency of residuals from observation i and i' to be correlated.

The logic of TMO is that while most pairs of units are not correlated, some will be, and the correlation between auxiliary outcomes $\hat{\rho}_{i,i'}$ can be used to determine those observations which are correlated across space. Specifically, setting some threshold value δ as the value above which observations are inferred to be meaningfully correlated, we can write $\hat{\Omega}$ as:

$$\hat{\Omega} = \begin{cases} \hat{\varepsilon}_i^0 \hat{\varepsilon}_{i'}^0 & \text{if } i = i', \\ \hat{\varepsilon}_i^0 \hat{\varepsilon}_{i'}^0 & \text{if } \hat{\rho}_{i,i'} \geq \delta, \\ 0 & \text{if } \hat{\rho}_{i,i'} < \delta. \end{cases}$$

In essence, this threshold term δ is used to indicate which units have a systematic correlation between residuals across space, and as such account for this correlation in variance estimates. Note that previously, off diagonal terms were $\hat{\varepsilon}_i^0 \hat{\varepsilon}_{i'}^0$, were allowed to be non-zero if they fulfilled specific *geographic* criteria such as sharing a cluster, whereas here with the incorporation of δ , these terms are allowed to be non-zero if *auxiliary outcomes themselves* are observed to be correlated across space.

DellaVigna et al. (2025) show that under two assumptions, these procedures result in a consistent variance estimate. Specifically, these assumptions are referred to as ‘proportionality’ and ‘sparsity’. The first of these implies that the covariance of residuals for auxiliary outcomes is informative of the correlation across residuals for the outcome of interest. And the second assumption implies that there is some limit on the number of units which are actually correlated with any other unit.

Determining Thresholds for Cross-Space Correlations Provided that such assumptions are reasonable, the remaining question is what threshold value for δ should be used as a

threshold for determining correlations. In effect, most pairs of observations will not be correlated (“null pairs”), while some small proportion (“non-nulls”) will be correlated. Selecting a threshold thus can be viewed as a trade-off between potentially excluding relevant non-nulls (which will understate the variance) and including irrelevant nulls (which will overstate the variance). DellaVigna et al. (2025) propose determining the optimal threshold δ^* as the value which sets the proportion of misclassification among pairs flagged as being ‘non-null’ at 50%. This is a False Discovery Rate (FDR), and implies that if δ^* were lowered, more than half of the pairs added would actually be null, whereas if δ^* were increased, more than half of the pairs removed would actually be non-null.

Empirically, the value δ^* is determined with this trade-off in mind. In order to resolve this trade-off, we define as $F_0(\cdot)$ the (unknown) distribution function of empirical correlations $\hat{\rho}_{i,i'}$ for **null** pairs of units (we discuss the estimation of this below). Similarly, we can define $F_0^+(\cdot) \equiv 2(1 - F_0(\cdot))$ as the right distribution for the *absolute* correlations $|\hat{\rho}_{i,i'}|$.³ Similarly, we denote as $F^+(\delta) = \Pr(|\hat{\rho}_{i,i'}| \geq \delta)$ the right distribution function for absolute correlations for both null and non-null pairs of units (*i.e.* the unconditional distribution).

We can write the probability that observations whose $\hat{\rho}_{i,i'} \geq \delta$ are non-null and null as follows for a specific threshold choice δ :

$$\Pr(\text{Non-Null} \mid |\hat{\rho}_{i,i'}| \geq \delta) = \frac{F^+(\delta) - p_0 F_0^+(\delta)}{F^+(\delta)} \quad \Pr(\text{Null} \mid |\hat{\rho}_{i,i'}| \geq \delta) = \frac{p_0 F_0^+(\delta)}{F^+(\delta)}, \quad (8)$$

where p_0 refers to the proportion of pairs of units with a null relationship across space. We can then take the difference between the probability of observations being null and non-null as proportional to the following:

$$Q(\delta) = (F^+(\delta) - p_0 F_0^+(\delta)) - p_0 F_0^+(\delta) = F^+(\delta) - 2p_0 F_0^+(\delta). \quad (9)$$

A key property to note with $Q(\delta)$ is that this will be increasing in δ if more nulls are admitted than non-nulls, and decreasing in δ if more non-nulls are admitted than nulls. Thus, maximizing $Q(\delta)$ with respects to δ finds the point at which half the added pairs are null, and half are non-null.

This suggests an (infeasible) criteria to select delta as maximizing the following quantity $\tilde{Q}$:

$$\tilde{Q}_{N,d}(\delta) = \hat{F}^+(\delta) - 2p_0 F_0^+(\delta) \quad \text{where} \quad \hat{F}^+(\delta) = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{i' < i} \mathbb{1}\{|\hat{\rho}_{i,i'}| > \delta\}. \quad (10)$$

This quantity cannot yet be estimated for two reasons. Firstly, while the distribution of both null and non-null pairs can be estimated from data as $\hat{F}^+(\delta)$, the distribution of just null pairs $\hat{F}_0^+(\delta)$ is unknown. And secondly, the proportion of nulls themselves (p_0) is

³Could point to an appendix where we simulate an example and show this in a step-by-step way so readers can easily understand the mechanics...?

also unknown. In line with the sparsity assumption, the quantity p_0 can (conservatively) be set equal to 1. This leads to a feasible criteria:

$$\hat{Q}_{N,d}(\delta) = \hat{F}^+(\delta) - 2F_0^+(\delta), \quad (11)$$

where the optimal threshold δ^* is chosen as the quantity that maximizes (11).

Thus, a final step required to actually estimate $\hat{Q}_{N,d}$ is to estimate the null distribution $F_0(\cdot)$. This cannot be estimated directly as the distribution of $\hat{\rho}_{i,i'}$ given that this distribution will contain both null and non-null pairs. However, given that most pairs are assumed to actually be null pairs, this leads to the proposal in DellaVigna et al. (2025) to estimate this null distribution by fitting a Normal distribution to the central part of the empirical distribution of $\hat{\rho}_{i,i'}$ which will overwhelmingly consist of null pairs. In the interests of completion, if $\Phi_v(\cdot)$ refers to the normal distribution with variance v , and $\Phi_v^{-1}(\cdot)$ its inverse, then the interquartile function can be written as $IQR_{v,0} = \Phi_v^{-1}(0.75) - \Phi_v^{-1}(0.25)$. Similarly, defining the empirical distribution of observed correlations as:

$$G_{N,d}(\delta) = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{i' < i} \mathbb{1}\{\hat{\rho}_{i,i'} < \delta\},$$

we can define the inverse and interquartile range of observed correlations (respectively) as $G_{N,d}^{-1}(\cdot)$ and $\widehat{IQR} = G_{N,d}^{-1}(0.75) - G_{N,d}^{-1}(0.25)$. The distribution F_0^+ is thus estimated by setting v to the quantity which resolves $IQR_{\hat{v},0} = \widehat{IQR}$.

At the beginning of Section 4 we will examine an empirical example which steps through each of these steps, in essence considering the estimation of $\hat{\rho}_{i,i'}$, $G_{N,d}(\delta)$, $\hat{v}$, and δ^* , and resultingly, the standard errors and confidence intervals corresponding to our regression estimate of interest $\hat{\tau}$. Examining this in practice helps to elucidate the steps in both thresholding, and estimating cross-space correlations. A summarised statement of the entire estimation algorithm as defined in DellaVigna et al. (2025) is provided as Algorithm 1 in an Appendix to this article.

2.2 Alternative Estimators

While the above treatment begins with a bivariate linear regression model in (1), the procedures extend directly to the estimation environments most commonly encountered in practice. With additional covariates or fixed effects, the residualised treatment $\tilde{W}_i$, that is the residual from regressing W_i on all other regressors, takes the place of W_i in the sandwich formula, by the Frisch–Waugh–Lovell theorem; the `tmo` command constructs $\tilde{W}_i$ internally from the command supplied in `cmd()`. With instrumental variables, the same logic applies to the first-stage projection: the command builds the appropriate 2SLS residualisation, so that the reported coefficient is the second-stage estimate while the variance is adjusted using TMO. In panel settings, residuals for each location are stacked across time periods, cross-location correlations are computed using all periods jointly, and the sandwich accumulates contributions over pairs of locations and pairs of

periods. Formal treatments of each extension are provided in [DellaVigna et al. \(2025, Appendix B.3\)](#). In terms of implementation, `tmo` currently supports `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2` and `ivreghdfe`, as documented in [Section 3](#).

2.3 Combining TMO with Distance-Based Procedures

TMO and the procedures reviewed above are not mutually exclusive. Each of the estimators discussed in this section can be written as a sandwich estimator in which the contribution of a pair of locations (i, i') enters with some weight $w_{ii'}$. Clustering sets $w_{ii'} = 1$ within clusters and 0 elsewhere; Conley-type estimators set $w_{ii'} = K(d_{ii'}/b)$, a kernel in distance; and SCPC implies a dense kernel $w_{ii'} = P_{ii'}$ derived from its low-frequency principal components, under which *every* pair receives some (typically small) weight.

This common structure makes combination natural. The combined estimator implemented in `tmo` retains the raw covariance term (weight 1) for every pair selected by the TMO threshold, and, in the clustering and uniform-distance cases, for every pair included by the baseline procedure; remaining pairs enter with the baseline weight:

$$\hat{\mathbb{V}}_{combined} \propto \sum_{(i, i') : |\hat{\rho}_{ii'}| \geq \hat{\delta}} \hat{\varepsilon}_i \hat{\varepsilon}_{i'} \tilde{W}_i \tilde{W}_{i'} + \sum_{\text{remaining pairs}} w_{ii'} \hat{\varepsilon}_i \hat{\varepsilon}_{i'} \tilde{W}_i \tilde{W}_{i'}, \quad (12)$$

where the first summation reflects pairs flagged by TMO (as well as own-cluster or within-cutoff pairs, where relevant), and $w_{ii'}$ in the second summation is the baseline weight: $\mathbb{1}\{d_{ii'} \leq b\}$ for a uniform distance kernel, $\max(1 - d_{ii'}/b, 0)$ for a Bartlett kernel (the spatial analogue of [Newey and West 1987](#)), or the SCPC kernel weight $P_{ii'}$. The intuition for such combinations is that the two sources of information are complementary: distance-based procedures are well suited to diffuse, geographically organised dependence, while TMO captures strong dependencies between particular pairs of locations, which need not be geographically close, revealed by the auxiliary outcomes. [Section 4](#) illustrates each combination, which is requested in `tmo` through the `cmd()` (clustering), `distthreshold()/distkernel()` (distance), and `scpc_cmd()` (SCPC) options.

3 The `tmo` command

3.1 Syntax

The `tmo` command is an e-class command with the following syntax:

```
tmo, cmd(cmdline) x(varname) ylist(varlist) idvar(varname) [options]
```

3.2 Options

`cmd(cmdline)` *cmdline* is the command that produces the regression of interest. `tmo` currently supports regressions using `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2`, or `ivreghdfe`.

`x(varname)` Regressor of interest in `cmd()` for which to estimate TMO standard errors. Only the standard error of this variable is adjusted.

`ylist(varlist)` List of auxiliary outcomes used to estimate the correlation structure.

`idvar(varname)` Location identifier variable; must be unique (within each time period for panel case).

`misslimit(#)` Limit for proportion of observations allowed to be missing for auxiliary outcomes. Auxiliary outcomes missing more than `misslimit` are not used. Must be between [0,1]; default is 0.1.

Panel Setting

`timevar(varname)` Time identifier variable; must be declared in the panel case.

Distance-based Settings

`latitude(varname)` Latitude variable in signed decimal degrees.

`longitude(varname)` Longitude variable in signed decimal degrees.

`distthreshold(#)` Distance threshold allowing arbitrary correlation between pairs of locations that are `distthreshold()` or closer. Distances are measured in kilometres by default; see the `miles` option. This combines `tmo` with the distance-based adjustment of [Conley \(1999\)](#), which uses a uniform kernel by default; see `distkernel()`. Requires `latitude()` and `longitude()`.

`miles` Measure distances for `distthreshold()` in miles rather than kilometres.

`distkernel(kernelname)` Kernel used to weight the pairs admitted by `distthreshold()`. *kernelname* may be `uniform` or `bartlett`; the default is `uniform`.

uniform gives full weight to every pair of locations within the cutoff.

bartlett gives full weight to the pairs selected by the TMO threshold, and weight $1 - d_{ii'}/b$ to the remaining pairs within the cutoff b , so that weight declines linearly with distance and reaches zero at the cutoff.

Requires `distthreshold()`. May not be combined with `scpc_cmd()`.

Saving Figures and Tables

filesuffix(*str*) Folder path and base filename for saving figures and results. Required for the `plot` and `save` options.

savedyad Save Stata data file with correlation and contribution to standard error for each location pair.

plotq Save plot of optimal threshold estimator.

plothist Save plot for histogram of correlations between locations.

plothistnbins(*#*) Number of bins for histogram of correlations (default 10000).

plotse Save plot for standard error estimates across thresholds.

saveplotseest Save Stata data file with standard error estimates across thresholds.

saveest Save the results returned in `r()` to a Stata data file.

Custom Threshold

threshold(*#*) Set a custom threshold instead of the optimal threshold obtained by the interquartile-range method. Must be between 0 and 1.

thresholdoff Turns off the TMO adjustment entirely, leaving only the baseline procedure requested through `distthreshold()` or the clustering in `cmd()`.

SCPC Options

scpc_cmd(*cmdline*) Regression to which the SCPC adjustment is applied. This combines `tmo` with the SCPC method of Müller and Watson (2022b): pairs selected by the TMO threshold enter the variance with their raw covariance, while all remaining pairs enter through the implicit SCPC kernel weight. Requires `latitude()` and `longitude()`, and requires that `scpc` be installed.

scpc_uncond Turns on the unconditional SCPC inference setting.

Returned Objects

`tmo` is an e-class command, and stores the following in `e()`:

<code>e(beta)</code>	point estimate for the variable declared in <code>x()</code>
<code>e(orig_se)</code>	standard error from the original command in <code>cmd()</code>
<code>e(tmo_se)</code>	TMO-adjusted standard error
<code>e(lb)</code> , <code>e(ub)</code>	bounds of the TMO-adjusted 95% confidence interval
<code>e(threshold)</code>	optimal (or user-set) threshold $\hat{\delta}$
<code>e(pct_ge_thres)</code>	% of off-diagonal pairs included in the variance
<code>e(pct_ge_thres_nocl)</code>	% of pairs over the threshold outside clusters/Conley
<code>e(N)</code> , <code>e(N_loc)</code> , <code>e(N_clust)</code>	observations, locations, and clusters
<code>e(T)</code>	time periods (panel case)
<code>e(N_outcomes)</code>	number of auxiliary outcomes used
<code>e(dof)</code>	degrees of freedom of the fitted null distribution
<code>e(finite_sample_dof)</code>	finite-sample degrees-of-freedom adjustment
<code>e(df_r)</code>	residual degrees of freedom
<code>e(r2)</code> , <code>e(r2_a)</code> , <code>e(rmse)</code>	fit statistics from the original command
<code>e(scpc_cv)</code>	SCPC critical value (when <code>scpc_cmd()</code> is used)

The matrices `e(b)` and `e(V)` contain the point estimate and TMO-adjusted variance for the variable declared in `x()`, facilitating the exportation of results with routines such as `estout`.

4.1 An Illustrative Example: OLS and US counties

```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d i.stfips, r) x(EDU_college_d) ///
> ylist('ylist') i(fips) plothist plotq file("figures/")
file figures/_qt.pdf saved as PDF format
file figures/_hist.png written in PNG format
```

Linear Regression with TMO Number of obs = 3,060

						R-squared	=	0.3434
						Adj R-squared	=	0.3434
						Root MSE	=	0.1409
PIN_persin-d	Coefficient	Std. err.	t	P> t	[95% conf. interval]			
EDU_colleg-d	.1745893	.0197081	8.86	0.000	.1359466	.213232		
						Optimal threshold		0.389
						% of off-diag in SE est.		0.497
						% >= threshold (excl. clusters/Conley)		0.497
						# outcomes		90.000
						Degrees of freedom		59.986

The upper panel reports the conventional OLS results. The lower block summarizes the TMO adjustment. The *Optimal threshold* $\hat{\delta}$ is a data-driven cutoff for the absolute correlations across locations; pairs with $|\hat{\rho}_{i,i'}| \geq \hat{\delta}$ are treated as correlated when constructing $\hat{\Sigma}^{(0)}(\hat{\delta})$. The *% of off-diag in SE est.* row reports the share of off-diagonal entries retained by this rule, here roughly 0.5% of all county pairs, which underscores the sparsity of the selection and shows the extent of cross-location dependence incorporated in the variance. When TMO augments clustering or Conley, this row also records the additional pairs contributed by the baseline method. In this case 90 auxiliary outcomes were used to learn the correlation structure; finally, the degrees of freedom reports the width of the normal reference used to set $\hat{\delta}$ (estimated from the interquartile range of Fisher- z correlations).

The figures document the choice of $\hat{\delta}$. Figure 2a shows the empirical distribution of Fisher- z correlations across locations together with the fitted $N(0, \hat{v})$ reference; the dashed vertical lines at $\pm \hat{\delta}$ indicate which pairs populate the off-diagonal of $\hat{\Sigma}^{(0)}(\hat{\delta})$. Figure 2b then plots the threshold objective $\hat{Q}_{n,d}(\delta)$; the dashed line marks its maximizer at $\hat{\delta}$ (larger δ yields a more conservative selection of pairs). The reported standard error is computed as $(W^\top W)^{-1}(W^\top \hat{\Sigma}^{(0)}(\hat{\delta})W)(W^\top W)^{-1}$.

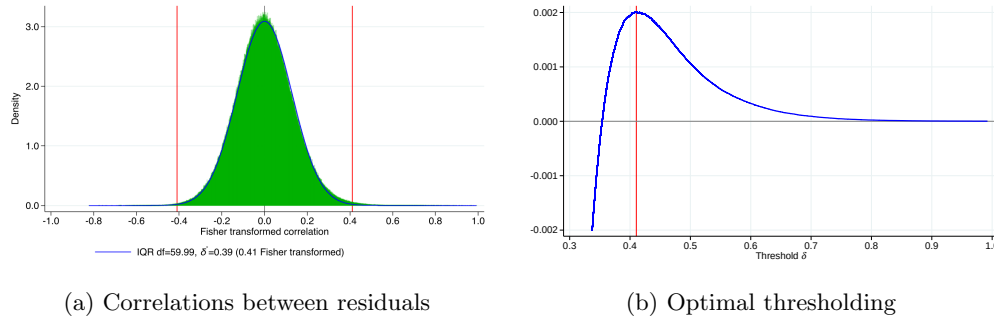


Figure 2: Estimation of the optimal threshold across US counties

4.2 Alternative Models

Instrumental Variables `tmo` can be used with instrumental-variables estimators. The example below is intended to illustrate the syntax rather than to support a causal claim: using the same cross-sectional dataset, we regress life expectancy on infant mortality, instrumenting `VST_infmort_d` with the median distance in miles to the nearest emergency, obstetrics and paediatric intensive care departments. Section 4.3 applies the command to a published instrumental-variables design. The `cmd()` option wraps the IV estimator that would be used in any case, and its contents follow the *native* syntax of the chosen command; `tmo` supports `ivreg2`, `ivreghdfe` and `ivregress`. The option `x()` marks the endogenous regressor whose standard error is to be adjusted, and `ylist()` supplies the auxiliary outcomes.

```
. tmo, cmd(ivreg2 life_d (VST_infmort_d = AHRQ_emerdist_d AHRQ_obgyndist_d ///
> AHRQ_pediadist_d)) x(VST_infmort_d) ylist('ylist') i(fips)
```

Linear Regression with TMO

Number of obs = 3,060
R-squared = -0.0464
Adj R-squared = -0.0464
Root MSE = 0.0460

life_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
VST_infmor-d	-.0082797	.015985	-0.52	0.604	-.0396097	.0230503

Optimal threshold	0.502
% of off-diag in SE est.	5.843
% >= threshold (excl. clusters/Conley)	5.843
# outcomes	87.000
Degrees of freedom	22.522

The output mirrors the cross sectional case: the reported coefficient equals the second-stage IV estimate from the underlying command, while the standard error is TMO-adjusted; the lower block reports the optimal threshold, the share of pairs re-

tained, and the remaining diagnostic quantities, all with the same interpretation as before.

Panel Data Panel-ready data are available as `county_panel.dta` in the project repository on GitHub.

In panel settings, `tmo` incorporates the time dimension through the unit and time identifiers specified in `idvar()` and `timevar()`. The command treats $\{i, t\}$ as a panel, reshapes the normalized residuals into an $N \times T$ array, and computes cross location correlations using all periods. The sandwich matrix is then built from location pairs (i, i') combined with time pairs, which increases computational cost; for large panels the program prints occasional progress messages (“Computed X% of sandwich”) as it accumulates contributions by blocks.

```
. tmo, cmd(reg EMN_farm EDU_publicenroll i.year i.stfips, cluster(fips)) ///
> x(EDU_publicenroll) ylist('ylist') i(fips) t(year)
Linear Regression with TMO
Number of obs = 6,222
R-squared      = 0.3426
Adj R-squared  = 0.3426
Root MSE      = 0.7683
```

EMN_farm	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
EDU_public-1	.1355818	.0758575	1.79	0.074	-.0131541	.2843177

```
Optimal threshold      0.545
% of off-diag in SE est. 0.892
% >= threshold (excl. clusters/Conley) 0.892
# outcomes              63.000
Degrees of freedom      24.612
```

The output is analogous to the cross sectional case. The effective sample size increases because the data expand over time, and the optimal threshold may differ since correlations are constructed using the panel information. All reported fields keep the same interpretation as in the cross sectional setting.

Other routines The regression of interest may be specified with any of the supported estimation commands, and equivalent specifications yield identical results regardless of the wrapper used. The log below estimates the same state-fixed-effects model three ways, using `regress` with factor-variable notation, `reghdfe` with `absorb()` and `areg`, and reports timings. The three TMO standard errors coincide; run-times differ only through the speed of the underlying estimation command, since the auxiliary-outcome regressions are re-estimated with the same wrapper.

```
. timer clear
. timer on 1
```

```

. qui tmo, cmd(regress PIN_persincpc_d EDU_college_d i.stfips, r) ///
> x(EDU_college_d) ylist(`ylist`) i(fips)
. timer off 1
.
. timer on 2
. qui tmo, cmd(reghdfe PIN_persincpc_d EDU_college_d, vce(r) abs(stfips)) ///
> x(EDU_college_d) ylist(`ylist`) i(fips)
. timer off 2
.
. timer on 3
. qui tmo, cmd(areg PIN_persincpc_d EDU_college_d, r abs(stfips)) ///
> x(EDU_college_d) ylist(`ylist`) i(fips)
. timer off 3
. timer list
1:      5.86 /      1 =      5.8590
2:      5.60 /      1 =      5.5970
3:      5.60 /      1 =      5.6000

```

4.3 Alternative settings

Combining tmo with other spatial dependence procedures

The `tmo` command can be combined with existing approaches to spatial inference. This is useful when the researcher wants to preserve a conventional restriction on the covariance matrix, such as arbitrary dependence within states, within a distance cutoff, or through the SCPC kernel, while allowing TMO to add further pairs whose auxiliary outcomes suggest residual dependence. Operationally, the original procedure defines a baseline set of pairs that enter the variance estimator. TMO then adds pairs with $|\hat{\rho}_{ii'}| \geq \hat{\delta}$, except where the baseline procedure already includes them.

TMO with clustered standard errors The simplest combination starts from the usual cluster-robust estimator. In the example below, the original regression clusters by state, and `tmo` is then called with the same clustered regression in `cmd()`. The command preserves the within-state covariance terms and uses the auxiliary outcomes to decide which cross-state pairs should also enter the variance.

```

. reg PIN_persincpc_d EDU_college_d, cluster(stfips)

```

Linear regression	Number of obs	=	3,060
	F(1, 47)	=	39.74
	Prob > F	=	0.0000
	R-squared	=	0.0708
	Root MSE	=	.16767
(Std. err. adjusted for 48 clusters in stfips)			

PIN_persin-d	Robust				
	Coefficient	std. err.	t	P> t	[95% conf. interval]

EDU_colleg-d	.2299788	.0364834	6.30	0.000	.1565835	.303374
_cons	1.934663	.0243343	79.50	0.000	1.885708	1.983617

```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d, cluster(stfips)) ///
> x(EDU_college_d) ylist('ylist') i(fips)
```

Linear Regression with TMO

Number of obs = 3,060
R-squared = 0.0705
Adj R-squared = 0.0705
Root MSE = 0.1677

PIN_persin-d	Coefficient	Std. err.	t	P> t	[95% conf. interval]
EDU_colleg-d	.2299788	.0381001	6.04	0.000	.1533312 .3066263

Optimal threshold 0.433
% of off-diag in SE est. 3.508
% >= threshold (excl. clusters/Conley) 0.438
outcomes 90.000
Degrees of freedom 45.192

TMO with Conley standard errors The most common distance-based correction is that of [Conley \(1999\)](#), which allows arbitrary correlation between locations closer together than a chosen bandwidth. It is requested by supplying county centroids through `latitude()` and `longitude()` together with the bandwidth through `distthreshold()`, and it can be used either on its own, by adding `thresholdoff`, or combined with TMO, which is the default. In the combined estimator a pair of counties enters the variance if it lies within the bandwidth, as under Conley, or if its auxiliary outcomes co-move, as under TMO. The bandwidth is measured in kilometres unless `miles` is specified. By default all pairs within the bandwidth receive full weight, which is the uniform kernel of the original proposal; `distkernel(bartlett)` instead lets the weight of the pairs that TMO has not selected decline linearly with distance.

```
. tmo, cmd(reg PIN_persincpc_d EDU_college_d, r) ///
> x(EDU_college_d) ylist('ylist') i(fips) lat(_CY) lon(_CX) ///
> distthreshold(150) miles
```

Linear Regression with TMO

Number of obs = 3,060
R-squared = 0.0705
Adj R-squared = 0.0705
Root MSE = 0.1677

PIN_persin-d	Coefficient	Std. err.	t	P> t	[95% conf. interval]
EDU_colleg-d	.2299788	.0382903	6.01	0.000	.1549014 .3050561

Optimal threshold 0.433
% of off-diag in SE est. 3.580
% >= threshold (excl. clusters/Conley) 0.410

```

# outcomes      90.000
Degrees of freedom 45.192

. tmo, cmd(reg PIN_persincpc_d EDU_college_d, r) ///
> x(EDU_college_d) ylist(`ylist`) i(fips) lat(_CY) lon(_CX) ///
> distthreshold(150) miles distkernel(bartlett)
Linear Regression with TMO
Number of obs = 3,060
R-squared      = 0.0705
Adj R-squared  = 0.0705
Root MSE      = 0.1677

```

PIN_persin-d	Coefficient	Std. err.	t	P> t	[95% conf. interval]
EDU_colleg-d	.2299788	.0323617	7.11	0.000	.1665259 .2934316

```

Optimal threshold      0.433
% of off-diag in SE est. 3.580
% >= threshold (excl. clusters/Conley) 0.410
# outcomes             90.000
Degrees of freedom      45.192

```

TMO with SCPC The combination with SCPC ([Müller and Watson 2022b](#)) requires no separate step: adding `scpc_cmd()` to the same call is enough. The option takes the regression to which SCPC should be applied, and `tmo` runs `scpc` internally on the coordinates already supplied through `latitude()` and `longitude()`, recovers its implicit pair-level kernel, and combines it with the TMO selection in a single variance estimate. Pairs selected by TMO enter with their raw residual covariance and every other pair enters through the SCPC kernel weight. Note that `scpc` must be installed, and that it supports a narrower set of estimation commands than `tmo` itself, so the regression passed to `scpc_cmd()` may need to be written with `regress` or `ivregress` even when `cmd()` uses another wrapper.

```

. reg PIN_persincpc_d EDU_college_d, r
Linear regression
Number of obs      = 3,060
F(1, 3058)         = 168.81
Prob > F           = 0.0000
R-squared          = 0.0708
Root MSE          = .16767

```

PIN_persin-d	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
EDU_colleg-d	.2299788	.0177005	12.99	0.000	.1952727 .2646848
_cons	1.934663	.0089404	216.40	0.000	1.917133 1.952192

```

. scpc, latlong
found 3060 observations / clusters and 2-dimensional locations in s_*
Computing distances on surface of sphere treating s_1 as latitude and s_2 as lo
> ngitude

```

SCPC optimal $q = 6$ for maximal average pairwise correlation = 0.030
 SCPC Inference for first 2 coefficients

	Coef	Std_Err	t	P> t	95% Conf
EDU_colleg-d	.2299788	.0253915	9.06	0.000	.1625168
_cons	1.934663	.0335045	57.74	0.000	1.843191
	Interval				
EDU_colleg-d	.2974407				
_cons	2.026134				

```

. rename (s_1 s_2) (_CY _CX)
.
. tmo, cmd(regress PIN_persincpc_d EDU_college_d, r) ///
> x(EDU_college_d) ylist(`ylist`) i(fips) lat(_CY) lon(_CX) ///
> scpc_cmd(reg PIN_persincpc_d EDU_college_d, r)

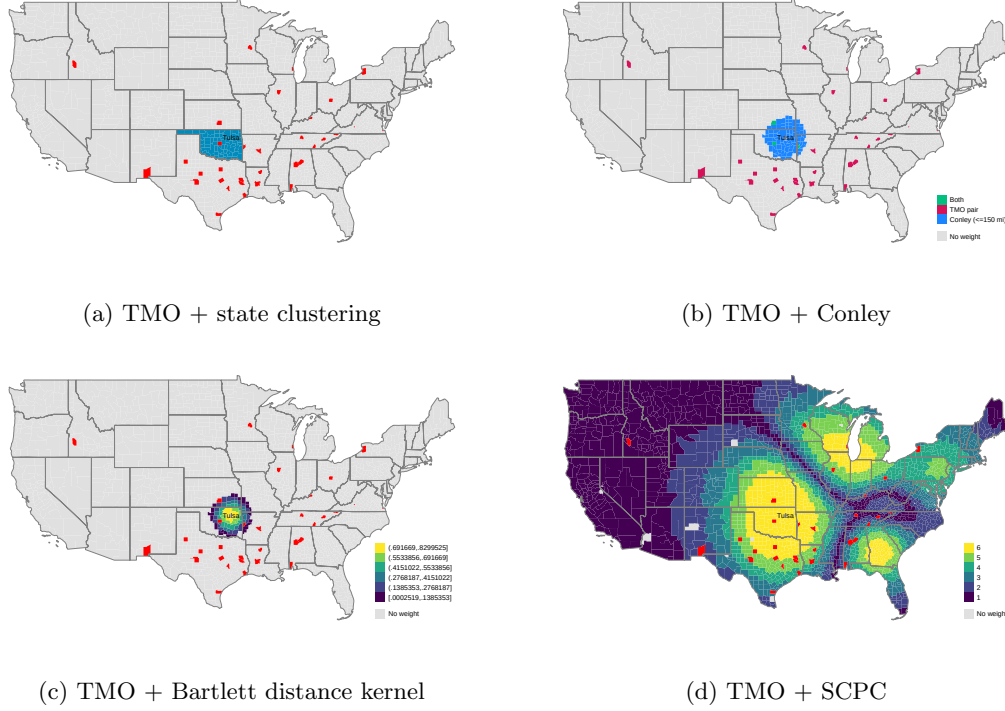
```

Linear Regression with TMO

Number of obs =	3,060
R-squared =	0.0705
Adj R-squared =	0.0705
Root MSE =	0.1677

PIN_persin-d	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
EDU_colleg-d	.2299788	.0287255	8.01	0.000	.1736556	.2863019

Optimal threshold	0.433
% of off-diag in SE est.	0.667
% >= threshold (excl. clusters/Conley)	0.667
# outcomes	90.000
Degrees of freedom	45.192



Notes: each panel is drawn from the pair-level output of the corresponding `tmo` run for the focal county of Figure 1. Panels b and c use a 150-mile bandwidth, the same cutoff as in Figure 1 and in the applications of Section 4.3, so that the shaded regions are directly comparable across figures.

Figure 3: Combining TMO with existing spatial-dependence adjustments

Figure 3 visualises each combination for the focal county of Figure 1, with every layer derived from the pair-level output of the corresponding `tmo` run. Two features deserve emphasis, since the combined estimators inherit very different structures from their two components. First, the TMO contribution is deliberately *sparse*. Only the small set of counties whose auxiliary outcomes systematically co-move with the focal county is selected, shown in red in panels c and d, and these counties may lie anywhere on the map. Second, the baseline contributions differ in their density. The Conley disk in panel b and the Bartlett kernel in panel c are *local and bounded*: weights are confined within the distance cutoff and are zero everywhere else, so that the map is mostly unshaded. The SCPC kernel, shown in the background of panel d, is instead *dense and global*. Because SCPC conducts inference through a handful of smooth low-frequency spatial components, every pair of counties receives some weight, without any threshold or cutoff, and the entire map is therefore shaded. This is not an artefact of the figure but a faithful depiction of the estimator: in the combined TMO–SCPC variance, the red counties contribute their raw residual covariance, while every remaining county contributes through the smoothly varying kernel weight shown in the background. The

comparison also clarifies the division of labour in equation (12). The baseline captures diffuse dependence that is organised by geography, sharply bounded in the case of Conley and smooth and global in the case of SCPC, while TMO adds the specific and potentially distant pairs that geography alone would miss.

A step-by-step guide for applied researchers

The examples so far document how to run the command. For a researcher considering TMO in their own work, the relevant questions are broader: whether an adjustment is needed at all, whether it should replace or complement a distance-based correction, and how to establish that the adjustment obtained is reliable. We suggest the following workflow:

1. **Start from the specification you would report anyway.** Nothing about the specification changes: the same regression, controls, fixed effects and level of clustering are passed to `cmd()`, and TMO adjusts its standard error. What matters at this stage is that the command be one `tmo` supports, and that the regressor whose standard error is of interest be named in `x()`.
2. **Build the collection of auxiliary outcomes.** This is the only genuinely new input the method requires, and the one that determines what the adjustment can detect. The criteria for assembling it are discussed below.
3. **Run `tmo` with the diagnostic output enabled** (`plothist`, `plotse` and `savedyad`). Passing the original command line, clustering included, means the existing inference is augmented rather than replaced.
4. **Decide whether the estimated correlation structure can be trusted.** The two diagnostic figures answer this: they show whether the null distribution against which pairs are selected is credible, and whether the reported adjustment is a stable feature of the data or an artefact of the particular threshold chosen. A failed check is a statement about the outcome collection, not about the threshold.
5. **Establish whether the detected dependence is organised by geography.** The pair-level file saved by `savedyad` shows which locations were selected and how far apart they are, and so indicates whether a distance-based correction would capture the same dependence.
6. **Compare the available corrections**, combining TMO with a distance-based or clustered baseline when dependence is partly geographic and partly not.
7. **Report the sensitivity of the adjustment to the outcome collection**, along with the outcome list itself.

Steps 2 and 4 carry most of the weight, and both are discussed at length by [DellaVigna et al. \(2025\)](#), whose recommendations we summarise here. On the collection of outcomes, the guiding trade-off is between the number of outcomes and their relevance.

Precision in the estimated correlations improves with the number of outcomes, and too few of them leave the procedure unable to separate genuine correlation from noise; but outcomes that do not share the spatial structure of the outcome of interest dilute the signal, and adding them can shrink the adjustment rather than sharpen it. Their evidence is that outcomes drawn from the same substantive area as the outcome of interest, and expressed in the same form, whether in levels or in differences, produce markedly stronger adjustments than an equally large collection of unrelated outcomes; and that variables taken from the study’s own replication package tend to outperform generic series collected from external sources. Their practical recommendation is to prefer the most relevant outcomes available subject to reaching at least twenty estimated degrees of freedom, a quantity `tmo` reports directly. Two requirements are dictated by the method rather than by relevance: the auxiliary outcomes must be measured for the same locations as the regression, and they must not measure the treatment, since correlations recovered from treatment-related outcomes reflect the spatial pattern of the treatment itself. Outcomes should also be brought to a common scale before use, following the cleaning protocol of DellaVigna et al. (2025, Appendix B.1): normalise variables that scale with population, take logarithms of positive levels, standardise within period, and winsorise the extreme tails, so that a single heavy-tailed variable cannot dominate the correlation estimates.

We illustrate each step using Bernini et al. (2023), a study of the effect of the Voting Rights Act on Black office-holding across 971 Southern counties. All auxiliary outcomes are drawn from the paper’s replication package, so every step below can be reproduced directly from publicly available materials.

Steps 1–3: baseline, outcomes, and the augmented run. The preferred specification (Table 2, Column 4 of Bernini et al.) regresses the 1964–1980 change in the share of Black elected officials on the interaction of the 1960 Black population share with federal coverage, together with the Black share, ten county controls and their coverage interactions, and state fixed effects, clustering by judicial division. Estimating this specification yields $\hat{\theta} = 0.103$, with a cluster-robust standard error of 0.045. For the auxiliary collection we follow the rules DellaVigna et al. (2025) state for this application: take all feasible variables in the replication package, then drop those that are missing for more than half of the estimation sample, those collinear with the right-hand side of the model, and those highly correlated with the main outcome, the regressor of interest, or the primary controls. The last rule requires a cutoff, which is not specified there; we exclude outcomes whose absolute correlation with any of those variables exceeds 0.8, and Table 3 below reports what happens when this rule is dropped or applied alone. With that cutoff the rules yield $d = 60$ auxiliary outcomes, matching the number used in the original analysis, and including, for example, the change in log gubernatorial turnout between 1940 and 1960 and the change in NAACP branches between 1942 and 1964. The exact list is not published, so the collection is ours rather than theirs, and the outcome-collection sensitivity in Step 7 is reported with that in mind. The log below runs `tmo` augmenting the original judicial-division clustering:


```

. tmo, cmd(reg ch_ShareBl_AllOfficials black_share60_lit_nc black_share60 ///
> c.urbanB60#literacy_nc c.unemp60#literacy_nc c.family_less_3000#literacy_
> nc ///
> c.pop60#literacy_nc c.school_low#literacy_nc c.cotton_suitability#literac
> y_nc ///
> c.cotton_share_land1964#literacy_nc c.anti_black_county#literacy_nc ///
> c.pro_black_county#literacy_nc c.rep_share_1964#literacy_nc ///
> ibn.STATE, nocon robust cluster(judicial_divisions_id)) ///
> x(black_share60_lit_nc) ylist(`ylist`) i(fips) misslimit(0.5) ///
> plothist plothistnbins(100) plotse file("figures/bernini")
file figures/bernini_hist.png written in PNG format
Calculating TMO SE over threshold at 0.00
Calculating TMO SE over threshold at 0.20
Calculating TMO SE over threshold at 0.40
Calculating TMO SE over threshold at 0.60
Calculating TMO SE over threshold at 0.80
Calculating TMO SE over threshold at 1.00
file figures/bernini_se_ratio_over_thres.pdf saved as PDF format
file figures/bernini_prop_above_thres.pdf saved as PDF format
Linear Regression with TMO
Number of obs = 971
R-squared = 0.5162
Adj R-squared = 0.5162
Root MSE = 5.1862

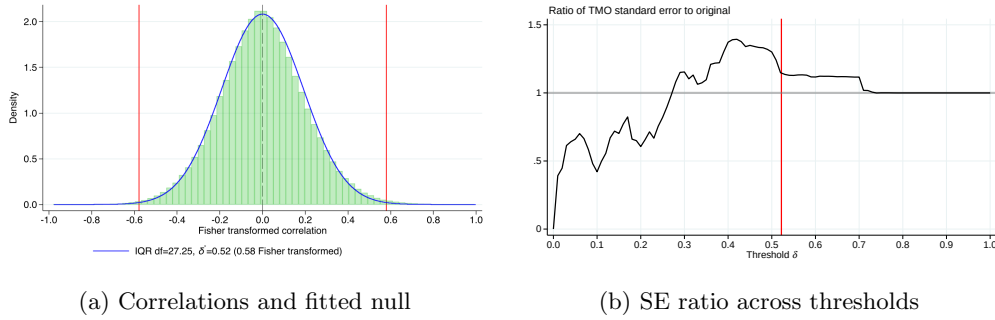
```

c~l0fficials	Coefficient	Std. err.	t	P> t	[95% conf. interval]
black_shar-c	.1029901	.0513034	2.01	0.047	.0014997 .2044805

Optimal threshold	0.522
% of off-diag in SE est.	1.632
% >= threshold (excl. clusters/Conley)	0.708
# outcomes	61.000
Degrees of freedom	27.253

Step 4: inspect the diagnostic figures. Two checks should be carried out before any TMO adjustment is used, based on the figures produced by `plothist` and `plotse` and displayed in Figure 4. In the correlation histogram (panel a), verify that: (i) the centre of the distribution is symmetric, unimodal, and closely tracked by the fitted Gaussian null, since the central mass is dominated by null pairs and a skewed or poorly-fitted centre indicates excessive common correlation across the auxiliary outcomes, which invalidates the estimated null; (ii) there is visible excess mass in the tails relative to the fitted null, which is the evidence of non-null pairs that the procedure is designed to capture; and (iii) the implied degrees of freedom ($\hat{df} = 1/\hat{v}$, reported in the output) reach at least 20, with larger values advisable in panel settings. In the threshold plot (panel b), verify that: (i) the curve is smooth and forms a plateau around $\hat{\delta}^*$, so that the reported adjustment does not depend on the exact threshold; (ii) the ratio tends to one as $\delta \rightarrow 1$, since with no pairs selected the baseline standard error must be recovered; and (iii) erratic movement is confined to low thresholds, where selection is dominated by noise pairs. Both checks pass here: the fit is adequate with clear tail excess and $\hat{df} = 27$,

above the recommended floor, and the curve is stable around $\hat{\delta}^* = 0.52$. When either check fails, the appropriate response is a larger or more relevant collection of outcomes, not a different threshold.



Notes: produced by the `plothist` and `plotse` options; the panels correspond to Figure 5 of DellaVigna et al. (2025).

Figure 4: Diagnostics for the application to Bernini et al. (2023)

Step 5: establish whether the dependence is geographic. This step determines the choice between corrections. Using the pair-level file saved by `savedyad`, merge in location characteristics and compute, among the pairs selected by the threshold, the share that are close in geographic distance and in other observable dimensions; comparison with the corresponding shares among all pairs indicates which dimensions organise the detected dependence. Table 1 reports this exercise, following the descriptive approach of Table 1 of DellaVigna et al. (2025). Geography is clearly at work: relative to all pairs, selected pairs are three times as likely to lie within 150 miles of one another and eight times as likely to share a judicial division. It is equally clear that geography is insufficient: fewer than a third of selected pairs are within the 150-mile band, similarity in population, poverty, or urbanisation adds little, and roughly 43% of the selected pairs are close on none of the criteria considered. Both a distance-based correction and TMO therefore have a role in this application.

	Among pairs with $ \hat{\rho}_{ii'} \geq \hat{\delta}^*$	Among all pairs
Within 150 miles	26.9%	9.0%
Same state	37.3%	11.8%
Same judicial division	7.3%	0.9%
Top 10% closest, 1960 population	10.3%	10.0%
Top 10% closest, 1960 poverty rate	13.0%	10.2%
Top 10% closest, 1960 urban share	11.2%	10.1%
Any of the above	56.3%	38.0%

Notes: share of county pairs satisfying each proximity criterion, among the pairs selected by the TMO threshold and among all county pairs; computed from the pair-level file saved by `savedyad`. County centroids are taken from the 2018 Census cartographic boundary files bundled with this article’s materials.

Table 1: Predictors of highly-correlated county pairs, Bernini et al. (2023) data

Step 6: compare and, where warranted, combine corrections. Table 2 compares the corrections directly. Consistent with Step 5, both channels matter here: the Conley correction alone raises the standard error by roughly 40%, and TMO contributes further pairs beyond the distance band. This application illustrates the general recommendation: when both geographic and non-geographic dependence are present, the combined estimator provides robustness to both.

	SE	Ratio to clustered
Cluster by judicial division	0.045	1.00
Conley (150 mi)	0.063	1.41 [9.0%]
TMO	0.051	1.15 [0.7%]
TMO + Conley (150 mi)	0.063	1.42 [9.6%]

Notes: ratios relative to the judicial-division-clustered standard error (0.045); brackets report the share of off-diagonal county pairs admitted to the variance.

Table 2: Method comparison, Bernini et al. (2023) application

Step 7: assess sensitivity to the outcome collection, and report it. The choice of auxiliary outcomes is the most consequential decision in the procedure, and errors are possible in both directions: with too few relevant outcomes, correlations are noisily estimated, the null distribution widens, and true non-null pairs are missed, while adding outcomes that do not share the correlation structure of the main outcome pulls the correlation estimates toward zero and dilutes the adjustment. The degrees-of-freedom statistic, which measures the effective number of independent outcomes rather than the raw count, is the relevant metric, together with the stability of the adjustment across reasonable collections. Table 3 reports the sensitivity exercise for this application: across alternative defensible implementations of the same selection rules, the diagnostics are essentially unchanged while the ratio spans 1.12–1.48. Because the adjustment inherits this sensitivity, we recommend publishing the auxiliary outcome list verbatim alongside

any TMO-adjusted result, as the code accompanying this article does, and reporting the adjustment’s range across reasonable collections rather than a single number.

Auxiliary collection	d	$\hat{df}$	$\hat{\delta}^*$	SE ratio
Correlation cutoff only, no family exclusions	57	26.1	0.50	1.48
Full selection rules (baseline)	60	27.3	0.52	1.15
Family exclusions only, no correlation cutoff	67	27.8	0.52	1.40
Baseline + regressor-family measures (misuse)	64	27.3	0.52	1.12

Notes: TMO applied to the Bernini et al. (2023) specification under alternative auxiliary collections. The fourth row deliberately adds three Black-population-share measures to illustrate an implementation error that the diagnostics do not flag (see below).

Table 3: Sensitivity of the TMO adjustment to the auxiliary outcome collection

Common implementation errors. Several implementation errors deserve attention; importantly, some are not detectable from the diagnostic figures and must instead be ruled out by design. First, *including measurements of the treatment among the auxiliary outcomes*. The misuse row of Table 3 adds three Black-population-share measures, which belong to the regressor’s family, to the collection: the reported diagnostics are essentially unchanged and the ratio appears unremarkable, yet the design is invalid, since dependence recovered from treatment-correlated outcomes partly reflects the spatial pattern of the treatment itself, biasing the adjustment in an unknown direction. Outcome selection is an identification choice, not a diagnostic one. Second, *operating at the degrees-of-freedom floor*: underpowered collections produce thresholds and adjustments that vary widely, including below one, across small changes to the outcome list (Section 4.3 provides an example); the sensitivity exercise of Table 3 should be reported rather than a single run. Third, *too few locations*: with samples in the low hundreds the null approximation degrades, and below roughly one hundred locations both diagnostic figures fail visibly, as in the 60-country re-analysis of Funke et al. (2023) reported by DellaVigna et al. (2025). Fourth, *unprepared outcome data*: auxiliary series should be brought to a common scale before use, following the cleaning protocol of DellaVigna et al. (2025, Appendix B.1), since a single heavy-tailed outcome can otherwise dominate the correlation estimates. Fifth, *pervasive spatial dependence*: the sparsity assumption requires that most pairs be approximately uncorrelated; in data with strong global trends of the spatial-unit-root type, no threshold separates null from non-null pairs, the histogram centre fails the Gaussian check, and trend removal must precede any correction.

Summary. The diagnostics map directly into practice. When the correlation-distribution and threshold-stability checks pass and the Step 5 exercise shows dependence concentrated at short distances, distance-based corrections are appropriate, and TMO should confirm rather than change them. When a substantial share of the detected dependence lies beyond any plausible distance kernel, distance-based corrections understate uncertainty and TMO is required; where both channels are present, as in the application

above, the combined estimator is the natural choice. When the diagnostics fail (few locations, a thin or irrelevant outcome collection, degrees of freedom at or below roughly 20, or instability of the adjustment across reasonable outcome sets), the appropriate conclusion is that the procedure cannot be applied reliably in that setting, and the auxiliary collection should be enriched before adjusted standard errors are reported.

A second application: instrumental variables in Bazzi et al. (2023)

A second application illustrates the workflow in an instrumental-variables design, where the choice of auxiliary outcomes requires one additional exclusion. Bazzi et al. (2023) study the long-run political consequences of the twentieth-century migration of Southern whites, in a cross-section of 1,886 U.S. counties. The main specification regresses the 2016 Trump vote share on the Southern-born white population share in 1940, instrumented by a shift-share variable built from 1900 migration networks and predicted migration flows, with historical baseline controls, state fixed effects, and standard errors clustered by 60×60 -mile grid cells (their Table 2, Panel A, Column 4, the main specification). Estimating this equation with `ivreghdfe` returns $\hat{\beta} = 1.026$ with a cluster-robust standard error of 0.167, matching the published values. The command is passed to `cmd()` exactly as written; `tmo` constructs the two-stage residualisation internally.⁴

Applying the Step 2 rules to the package yields $d = 85$ auxiliary outcomes, among them Republican vote shares in earlier elections, the Southern Black population share, and census, farm and manufacturing variables. The IV setting adds one requirement: the exclusions must cover the instrument's ingredients as well as the regressor's. Here the shift-share machinery, meaning the origin-state migrant stocks, the predicted flows and the associated truncation indicators, measures the treatment, and all of it is excluded by the same logic that excludes the regressor family.

```
. tmo, cmd(ivreghdfe Trump_share (pct_Southerners_white = iv_mig_hat1_00_2) ///
> `ctrl', cluster(km_grid_cel_code) a(statefip dummy_trunmig_hat1_00_2)) ///
> /
> x(pct_Southerners_white) ylist(`ylist') i(fips) misslimit(0.5) ///
> plothist plothistnbins(100) plotse file("figures/bazzi")
(file figures/bazzi_hist.png not found)
file figures/bazzi_hist.png written in PNG format
Calculating TMO SE over threshold at 0.00
Calculating TMO SE over threshold at 0.20
Calculating TMO SE over threshold at 0.40
Calculating TMO SE over threshold at 0.60
Calculating TMO SE over threshold at 0.80
Calculating TMO SE over threshold at 1.00
file figures/bazzi_se_ratio_over_thres.pdf saved as PDF format
file figures/bazzi_prop_above_thres.pdf saved as PDF format
Linear Regression with TMO
```

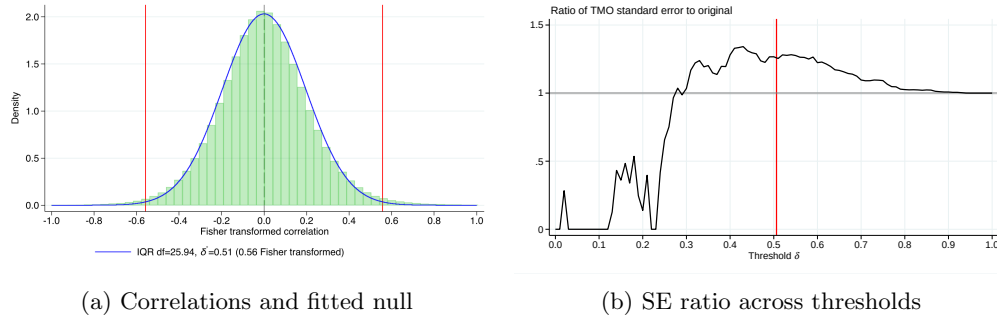
Number of obs	= 1,886
R-squared	= 0.4131
Adj R-squared	= 0.4131
Root MSE	= 8.7062

⁴The dataset is rebuilt from the raw materials in the paper's replication package; a note accompanying the file in the repository documents the construction. One endogenous regressor is supported.

Trump_share	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
pct_Southe-e	1.025748	.2086485	4.92	0.000	.6150292	1.436467

Optimal threshold	0.506
% of off-diag in SE est.	2.449
% >= threshold (excl. clusters/Conley)	1.797
# outcomes	86.000
Degrees of freedom	25.940

The Step 4 checks pass: the estimated degrees of freedom are 26, above the recommended floor, and the ratio-over-threshold curve is stable around the optimum $\hat{\delta}^* = 0.51$ (Figure 5).



Notes: produced by the `plottist` and `plotse` options.

Figure 5: Diagnostics for the application to Bazzi et al. (2023)

The Step 5 exercise, reported in Table 4, shows dependence that is strongly, though incompletely, organised by geography, much as in the county-level guide application. Selected pairs are six times more likely than average to lie within 150 miles of one another and nine times more likely to share a state, yet the proximity criteria together account for under half of the selected pairs. The most strongly correlated distant pairs connect opposite ends of the country: Aroostook County, Maine with Santa Clara County, California ($\hat{\rho} = 0.53$ at roughly 2,700 miles), for example, a pair no distance kernel would admit.

	Among pairs with $ \hat{\rho}_{ii'} \geq \hat{\delta}^*$	Among all pairs
Within 150 miles	26.3%	4.5%
Same 60×60 mi grid cell	5.4%	0.7%
Same state	33.8%	3.9%
Top 10% closest, population	15.9%	10.0%
Any of the above	47.0%	15.4%

Notes: share of county pairs satisfying each proximity criterion, among the pairs selected by the TMO threshold ($\hat{\delta}^* = 0.51$) and among all county pairs. Computed from the pair-level file saved by `savedyad`; the accompanying code documents the construction.

Table 4: Predictors of highly-correlated county pairs, Bazzi et al. (2023) data

Table 5 reports the Step 6 comparison. Consistent with Step 5, the distance-based corrections now move: Conley standard errors with a 150-mile bandwidth raise the standard error by 41%, close to the TMO-with-state-clusters adjustment. TMO combined with the original grid clustering yields a ratio of 1.25, and augmenting TMO with the Conley disk or with SCPC adds the residual distance-organised dependence, with combined ratios of 1.44–1.47. Under every adjustment the effect of Southern white migration remains statistically significant. Because `scpc` supports `ivregress`, the SCPC rows pass the specification to `scpc_cmd()` rewritten in `ivregress 2sls` form with explicit state dummies, which leaves the point estimates unchanged.

	SE	Ratio to clustered
Cluster by 60×60 mi grid	0.167	1.00
TMO	0.205	1.22 [1.9%]
TMO + grid clustering	0.209	1.25 [2.4%]
TMO + state clustering	0.241	1.44 [5.2%]
Conley (150 mi)	0.235	1.41
TMO + Conley (150 mi)	0.240	1.44 [6.2%]
SCPC	0.203	1.21
TMO + SCPC	0.246	1.47

Notes: ratios relative to the grid-clustered standard error (0.167); brackets report the share of off-diagonal county pairs admitted to the variance. The SCPC rows do not report this share, as SCPC weights all pairs through a smooth kernel rather than zeroing terms out.

Table 5: Method comparison, Bazzi et al. (2023) application

Finally, the Step 7 exercise in Table 6 probes the collection choice in both directions. Trimming the collection to $d = 60$, by additionally excluding the secondary control sets of the paper’s own Table 2 (the 1900 controls and sorting confounds) and the raw vote counts, drops the estimated degrees of freedom to 16, below the recommended floor. The resulting ratio happens to be similar here, but the diagnostics correctly flag the trimmed collection as underpowered, and the advice of Section 4.3 applies: prefer the broader defensible collection. The misuse row adds four measures from the regressor’s

family; as in the guide application, the reported diagnostics look unremarkable, which is precisely why treatment measures must be excluded by design rather than by inspection.

Auxiliary collection	d	$\hat{df}$	$\hat{\delta}^*$	SE ratio
Package, exclusion rules (baseline)	85	25.9	0.51	1.25
Also excluding secondary control sets	60	15.8	0.64	1.27
Package + regressor family (misuse)	89	27.8	0.49	1.21

Notes: TMO applied to the Bazzi et al. (2023) specification under alternative auxiliary collections, augmenting the original grid clustering. d is the number of outcomes used, $\hat{df}$ the estimated degrees of freedom of the null, and the SE ratio is relative to the grid-clustered standard error. The third row deliberately adds four Southern-white-share measures to illustrate an implementation error that the diagnostics do not flag (see Section 4.3).

Table 6: Sensitivity of the TMO adjustment to the auxiliary outcome collection

5 Discussion and Conclusion

In this article we have reviewed the main procedures available to Stata users for conducting inference in the presence of spatial dependence, and introduced the `tmo` command, which implements the Thresholding Multiple Outcomes procedure of DellaVigna et al. (2025). Rather than positing where spatial correlation lies, whether inside administrative borders or within a chosen distance, TMO estimates the location of dependence from auxiliary outcomes observed for the same units, and then permits arbitrary correlation among the (typically very small) set of location pairs whose outcomes systematically co-move.

Several practical lessons emerge from the examples documented here. First, the procedure is undemanding in terms of syntax: a researcher who can estimate their model with `regress`, `areg`, `reghdfe`, `ivregress`, `ivreg2` or `ivreghdfe` can obtain TMO-adjusted standard errors by wrapping the same command line in `cmd()`. Second, the quality of the adjustment rests on the auxiliary outcomes: the proportionality assumption requires that the available outcomes span the relevant dimensions of cross-location dependence, and the estimated degrees of freedom reported by the command provides a useful diagnostic of how informative the collection is. Researchers should favour rich collections of outcomes measured for all units. Third, TMO composes naturally with existing practice: it can augment clustered, distance-based, and SCPC inference, adding precisely those pairs of locations that the baseline procedure’s geography would overlook. In the county-level application considered here, these combinations behave as theory suggests: distance-based baselines capture local dependence, SCPC captures smooth global dependence, and TMO contributes strong, specific dependencies among pairs that may be far apart.

Finally, we note two caveats. TMO is designed for settings where dependence is *sparse*: most pairs of locations are (approximately) uncorrelated, with meaningful dependence concentrated in relatively few pairs. In data exhibiting very strong, pervasive

spatial dependence, of the form studied by Müller and Watson's recent work on spatial unit roots, sparsity fails by construction, and researchers should first diagnose and remove such trends before applying TMO or, indeed, most other spatial corrections. And as with all procedures discussed here, the number of auxiliary outcomes and locations drives computational cost; the command is written to remain practical at the scale of U.S. counties on a standard laptop.

Acknowledgments

We are grateful to ...

6 References

- Becker, S. O., P. D. Boll, and H.-J. Voth. 2026. Testing and Correcting for Spatial Unit Roots in Regression Analysis. *Stata Journal* 26(2): 177–202.
- Cameron, A. C., and D. L. Miller. 2015. A Practitioner’s Guide to Cluster-Robust Inference. *Journal of Human Resources* 50(2): 317–372.
- Conley, T. 1999. GMM estimation with cross sectional dependence. *Journal of Econometrics* 92(1): 1–45.
- DellaVigna, S., G. Imbens, W. Kim, and D. M. Ritzwoller. 2025. Using Multiple Outcomes to Adjust Standard Errors for Spatial Correlation. <https://arxiv.org/abs/2504.13295>.
- Eicker, F. 1967. Limit theorems for regressions with unequal and dependent errors. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, 59–82. University of California Press.
- Huber, P. J. 1967. The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, 221–233. University of California Press.
- Liang, K.-Y., and S. L. Zeger. 1986. Longitudinal data analysis using generalized linear models. *Biometrika* 73(1): 13–22.
- Long, J. S., and L. H. Ervin. 2000. Using Heteroscedasticity Consistent Standard Errors in the Linear Regression Model. *The American Statistician* 54(3): 217–224.
- MacKinnon, J. G., and H. White. 1985. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics* 29(3): 305–325.
- Moulton, B. R. 1986. Random group effects and the precision of regression estimates. *Journal of Econometrics* 32: 385–397.
- . 1990. An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units. *The Review of Economics and Statistics* 72(2): 334–338.
- Müller, U. K., and M. W. Watson. 2022a. Spatial Correlation Robust Inference in Linear Regression and Panel Models. *Journal of Business & Economic Statistics* 41(4): 1050–1064.
- . 2022b. Spatial Correlation Robust Inference. *Econometrica* 90(6): 2901–2935.

- Newey, W. K., and K. D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55: 703–708.
- Rogers, W. H. 1993. sg17: Regression standard errors in clustered samples. *Stata Technical Bulletin* 13: 19–23. Reprinted in *Stata Technical Bulletin Reprints*, Vol. 3, pp. 88–94.
- White, H. 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica* 48: 817–838.

About the authors

Stefano DellaVigna is . . .

Guido Imbens is the Applied Econometrics Professor and Professor of Economics at Stanford Graduate School of Business.

Woojin Kim is. . .

David M. Ritzwoller is . . .

Damian Clarke is an Associate Professor at The Department of Economics of The Universidad de Chile and the University of Exeter, a Research Fellow at IZA and an Associate at the Millennium Institute for Market Imperfections and Public Policy.

Nicolás París is. . .

Appendix

Algorithm 1 Thresholding Multiple Outcomes (TMO)

Require: Outcome $Y_i^{(0)}$ and treatment W_i for units $i = 1, \dots, n$

Ensure: Estimate of the variance of $\hat{\tau}^{(0)}$, the least-squares regression coefficient of $Y_i^{(0)}$ on W_i

- 1: Identify a collection of auxiliary outcomes $Y_i^{(1)}, \dots, Y_i^{(d)}$.
- 2: For each unit i and outcome j , compute the normalized residuals $\hat{\varepsilon}_i^{(j)}$
- 3: For each pair of units $i \neq i'$, measure the correlation $\hat{\rho}_{i,i'}$
- 4: Approximate the null distribution of $\hat{\rho}_{i,i'}$ with $N(0, \hat{v})$, where $\hat{v}$ solves

$$\Phi_{\hat{v}}^{-1}(0.75) - \Phi_{\hat{v}}^{-1}(0.25) = G_{n,d}^{-1}(0.75) - G_{n,d}^{-1}(0.25),$$

where $\Phi_v(\cdot)$ is the CDF of $N(0, v)$, and $G_{n,d}(\cdot)$ is defined (paper)

- 5: Choose the threshold $\hat{\delta}^*$ by maximizing

$$\hat{Q}_{n,d}(\delta) = (\hat{F}_{n,d}(\delta) - F_0^+(\delta)) - F_0^+(\delta), \quad \delta > 0,$$

where $F_0^+(\delta) = 2(1 - \Phi_{\hat{v}}(\cdot))$ and $\hat{F}_{n,d}(\cdot)$ defined in (paper).

- 6: Collect the estimates

$$\hat{\sigma}_{i,i'}^{(0)}(\hat{\delta}^*) = \begin{cases} \hat{\varepsilon}_i^{(0)} \hat{\varepsilon}_{i'}^{(0)}, & i = i', \\ \hat{\varepsilon}_i^{(0)} \hat{\varepsilon}_{i'}^{(0)} \mathbb{I}\{|\hat{\rho}_{i,i'}| \geq \hat{\delta}^*\}, & i \neq i', \end{cases}$$

into the matrix $\hat{\Sigma}^{(0)}(\hat{\delta}^*)$.

- 7: **Return** the variance estimate

$$\hat{V}(\hat{\delta}^*) = (W^\top W)^{-1} (W^\top \hat{\Sigma}^{(0)}(\hat{\delta}^*) W) (W^\top W)^{-1}.$$
